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IERG4320 Data Science in Practice (2025-26)

Assignment 4

Expected time: 6 hours

Learning outcomes:
1. To practise data visualization using Python. 2. To practise natural language processing using Python.

Instructions:

1. Do your own work. You are welcome to discuss the problems with your fellow classmates. Sharing ideas is great, and do write your own explanations.
2. If you use help from the AI tools, e.g. ChatGPT, DeepSeek and Gemini, write clearly how much you obtain help from the AI tools. No marks will be taken away for using any AI tools with a clear declaration.
3. All work should be submitted onto the blackboard before the due date.
4. You are advised to submit the following two items.
    a. A .pdf file containing the answers of the written part.
    b. A Assignment4_1155xxxxxx.py file storing all your programs for the five problems. (1155xxxxxx refers to your student ID)
5. Do type/write your work neatly. If we cannot read your work, we cannot grade your work.
6. We will grade your work based on Anaconda Python version 1.13.
7. The sample output in the specification is run based on Windows. Your output may be different if you run the programs on Mac or Linux machines.
8. If you do not put down your name, student ID in your submission (both .pdf and .py), you will receive a 10% mark penalty out of the assignment 4.
9. Due date:
    a. Session A: TBC January, 2026 (Thursday) 23:59
    b. Session B: TBC April, 2026 (Saturday) 23:59

Short questions (25%)
Answer all questions.

1. You are working as a senior data analyst in the Highway Department. You are building and improving the Autotoll Smart Solutions, charging the drivers automatically when driving through the tunnel.

    a. In Autotoll Smart Solutions, suggest **four** different input data. (4%)

License plate image,
Timestamp of passage,
Vehicle image for size, color and shape,
Sensor Data (like weight)

    b. In recognizing the identity of the car, what is the type of the data analytics? What is the type of the machine learning problem? Explain your answer. (4%)

type of the data analytics: Predictive Analytics
type of the machine learning problem: Supervised Learning-Classification
Explain: Labeled data, like images of license plates using corresponding characters, train model, it can predict the class, like specific character or number, for each part of a new and unseen license plate.

    c. Suggest **three** different machine learning models that can be used to recognize the identity of the car. (3%)

Support Vector Machine (SVM),
Random Forest,
Artificial Neural Network (ANN)

    d. In recognizing the identity of the car, suggest **two** metrics on how you can measure the performance of the model. Explain the calculation briefly. (4%)

1. Accuracy:
A straight-forward metric for classification is accuracy, which is the proportion of correctly identified characters out of the total number of characters. Its calculation is (True Positives+True Negatives)/(Total Predictions) or number of correct predictions / total number of predictions.
2. Precision:

For specific character like A, it can measure how many times the model is correct. when the model predicted A

$$Precision = \frac{TP}{TP+FP}$$

TP: True Positives, FP: False Positives

e. In recognizing the identity of the car, suggest **two** types of charts on how you can visualize the performance of the model. Explain the type of the chart and the axes labels briefly. (4%)

1. Confusion Matrix:
Chart Type**:** it is a matrix table that every row represent the actual class's instances, every column represent predicted class's instances.
Axes: X axis is Predicted Labels, Y axis is True Labels
2. ROC Curve:
Chart Type**:** it is a graph that performance of a classification model are shown at all classification thresholds, the key metric is the Area Under the Curve (AUC).
Axes: X axis is False Positive Rate (FPR), Y axis is True Positive Rate (TPR)

f. The fee depends on the type of the vehicle. Suggest **two** different ways in determining the types of the vehicle. (2%)

1. Image Classification: Machine learning model like ANN and CNN is trained by labeled dataset of vehicle images like cars, buses and trucks, it can classify new vehicle images.
2. Rule-Based System: Physical sensors are used to measure attributes like weight, height or number of axles, set of rules are applied to determine the vehicle type.

g. Your junior analyst mentioned that he wishes to perform a 10-fold cross validation for the recognition task. He divides the dataset into 10 folds, trains a model based on 9 folds and calculates the accuracy based on the left-out fold. At the end, the 10-fold cross validation accuracy is the average of the 10 intermediate accuracy. Is he correct? Explain your answer. (4%)

Yes, the junior analyst is correct. 10-fold cross validation can robustly evaluate model performance, it split dataset into 10 equal folds, which is trained 10 times. Each iteration uses 1 fold to be test set, uses other 9 folds to be training set. Final performance metric like accuracy will be matric from 10 iterations' average. To the model performance on unseen data, it is more stable and reliable estimate.